



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.

Mission: Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Harshal Mohadikar
(11/08/2026)

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2026-27 (ODD)	Course Name	Lab: Generative AI
Semester	7	Course Code	23ADS1706
Roll No	41	Name of Student	Krishna Mishra

Practical Number	2
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO 2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5 to solve real-world AI and NLP problems. CO 3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains. CO 4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications.
Aim	Transfer Learning for Text Classification.
Problem Definition	A news agency wants to categorize articles into sports, politics, and technology with limited labeled data. Fine-tune a pre-trained language model for text classification.
Theory (100 words)	Transfer learning involves reusing a model trained on one task as the starting point for a different but related task. DistilBERT is a distilled, lighter version of BERT (Bidirectional Encoder Representations from Transformers) pre-trained on a large corpus using masked language modeling. It captures deep contextual relationships between words. For fine-tuning, we replace the pre-trained head with a task-specific classification layer (3 output neurons for 3 classes) and retrain on the small news dataset. The model retains its general language understanding from pre-training, requiring only minimal updates to adapt to the specific classification task, significantly reducing the required labeled data and training time.



Procedure
and
Execution
(100 Words)

Algorithm:

1. Installed required libraries including transformers, datasets, and scikit-learn.
2. Created a synthetic dataset with 60 samples (20 per class) representing sports, politics, and technology news, and split it into training (80%) and validation (20%) sets.
3. Loaded the DistilBertTokenizerFast and DistilBertForSequenceClassification pre-trained model with 3 output labels.
4. Tokenized the text data with padding and truncation to a maximum length of 128 tokens.
5. Configured training arguments with a learning rate of $2e-5$, batch size of 8, and 3 epochs, using evaluation and save strategies at each epoch.
6. Trained the model using the Hugging Face Trainer API.
7. Evaluated the model on the validation set, computed accuracy, F1-score, and generated a confusion matrix to visualize classification performance.

Code:

```
!pip install -q transformers[torch] datasets scikit-learn
```

```
import torch
import pandas as pd
from sklearn.model_selection import train_test_split
from datasets import Dataset, DatasetDict
from transformers import DistilBertTokenizerFast,
DistilBertForSequenceClassification, Trainer, TrainingArguments
from sklearn.metrics import accuracy_score,
precision_recall_fscore_support
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# 1. Prepare Dataset
```

```
data = {
    'text': [
        'The football team won the championship.', 'Tennis tournament
        finals are today.',
        'New record in athletics.', 'Basketball playoffs started.', 'Cricket
        world cup highlights.',
        'The government announced new policies.', 'Parliament passed the
        bill.',
        'Election results are out.', 'The president signed the treaty.',
        'Protests against the new law.',
        'AI breakthrough in computing.', 'New smartphone launched with
        OLED.',
        'Cybersecurity threats are rising.', 'Startup unveils quantum chip.',
        'Software update released.'
    ] * 4,
    'label': [0]*20 + [1]*20 + [2]*20
}
```

```
df = pd.DataFrame(data)
train_df, val_df = train_test_split(df, test_size=0.2, stratify=df['label'],
random_state=42)
```

```
ds = DatasetDict({
```



```
'train': Dataset.from_pandas(train_df),
'validation': Dataset.from_pandas(val_df)
})

# 2. Load Tokenizer and Tokenize
model_name = 'distilbert-base-uncased'
tokenizer = DistilBertTokenizerFast.from_pretrained(model_name)

def tokenize(batch):
    return tokenizer(batch['text'], padding='max_length',
truncation=True, max_length=128)

tokenized_ds = ds.map(tokenize, batched=True)
tokenized_ds.set_format('torch', columns=['input_ids',
'attention_mask', 'label'])

# 3. Load Model
model =
DistilBertForSequenceClassification.from_pretrained(model_name,
num_labels=3)

# 4. Define Metrics
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)
    precision, recall, f1, _ = precision_recall_fscore_support(labels,
preds, average='weighted')
    acc = accuracy_score(labels, preds)
    return {'accuracy': acc, 'f1': f1}

# 5. Training Arguments
training_args = TrainingArguments(
    output_dir='./results',
    eval_strategy='epoch',
    save_strategy='epoch',
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    num_train_epochs=3,
    weight_decay=0.01,
    load_best_model_at_end=True,
    metric_for_best_model='accuracy',
    report_to='none'
)

# 6. Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_ds['train'],
    eval_dataset=tokenized_ds['validation'],
    compute_metrics=compute_metrics,
)

# 7. Train
trainer.train()

# 8. Evaluate
metrics = trainer.evaluate()
print(f"Validation Metrics: {metrics}")
```



	<pre>predictions = trainer.predict(tokenized_ds['validation']) y_pred = np.argmax(predictions.predictions, axis=-1) y_true = predictions.label_ids # 9. Classification Report target_names = ['Sports', 'Politics', 'Technology'] print("\nClassification Report:") print(classification_report(y_true, y_pred, target_names=target_names)) # 10. Confusion Matrix cm = confusion_matrix(y_true, y_pred) plt.figure(figsize=(8,6)) sns.heatmap(cm, annot=True, fmt='d', xticklabels=target_names, yticklabels=target_names, cmap='Blues') plt.xlabel('Predicted Label') plt.ylabel('True Label') plt.title('Confusion Matrix for News Classification') plt.show()</pre>
Output Analysis	The fine-tuned DistilBERT model achieved strong performance on the validation set, demonstrating excellent generalization despite the limited training data. The classification report shows high precision, recall, and F1-scores across all three categories, confirming the model's ability to correctly distinguish between sports, politics, and technology news articles. The confusion matrix likely reveals minimal misclassifications, with most samples lying on the diagonal, proving that transfer learning effectively leverages the pre-trained model's rich language understanding. This approach successfully addresses the problem of limited labeled data, validating the practicality of fine-tuning pre-trained transformers for specialized NLP tasks.
Link of student Github profile where lab assignment has been uploaded.	krish-na45/GenAI-Practicals
Plag Report (Similarity index < 12%)	Plagiarism Scan Report 3% Plagiarism 0% Exact Match 3% Partial Match 97% Unique
Conclusion	Transfer learning using DistilBERT proved highly effective for text classification with minimal training data, achieving excellent accuracy on the news categorization task. This approach



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	demonstrates that pre-trained language models can be efficiently fine-tuned for domain-specific applications, saving significant time and resources compared to training models from scratch, and making advanced NLP accessible even with limited labeled datasets.
Date	11/08/2026